Prepared for submission to JHEP

An exact degree-ten classification of local invariants of the ten-dimensional self-dual five-form

AUTHOR LIST – HUMAN ACTION REQUIRED[a]

[a]*AFFILIATIONS – HUMAN ACTION REQUIRED*

E-mail: corresponding.author@placeholder.invalid

Abstract: The self-dual five-form field strength in ten dimensions carries a large ring of Lorentz invariants, of which only a handful have been written down explicitly. We determine the degree-ten graded piece of that ring exactly. Working over finite fields with certified linear algebra, we construct a fourteen-dimensional atlas of degree-ten invariants and resolve its internal structure completely: the two-dimensional subspace of products of lower-degree invariants, the twelve-dimensional span of the published candidate structures, the twelve-dimensional span of the graph generators, and the eleven-dimensional subspace reachable by the stress-tensor flow that motivates these theories. The resulting quotient is three-dimensional, and we give explicit compact representatives for it. We show that two decompositions which both read twelve plus two are not the same decomposition: the published span contains one product direction, so its non-product content is eleven. Independently, we implement a bridge between the tensor description and a spinor-variable description, fixing every convention explicitly and verifying the map is an isomorphism onto the gamma-traceless symmetric chiral square. The two descriptions are then evaluated at a common registry of five-forms and their invariant spans compared directly. We also replace the floating-point Jacobian rank computation used previously by an exact modular one over the complete candidate selection. Because the coordinate basis is integral, the resulting rank is an unconditional lower bound in characteristic zero rather than a tolerance-dependent estimate, and it is witnessed by an explicit non-vanishing minor; it meets the analytic count of eighty-one from below.

Keywords: Extended Supersymmetry, p-branes, Differential and Algebraic Geometry

Contents

Draft status. This is a private submission candidate. The author list, affiliations, acknowledgments, funding, licence and data DOI are unresolved placeholders and are listed in `submission_candidate/AUTHORSHIP_DECISION_REQUIRED.md`. The physics interpretation of section 12 and the novelty statements require coauthor confirmation. Not for distribution.

1 Introduction

A self-dual five-form field strength $F_{\mu_1\dots\mu_5}$ in ten Lorentzian dimensions has 126 independent components. The Lorentz group $SO(1,9)$ has dimension 45 and acts with trivial generic stabiliser, so a generic orbit is 45-dimensional and the number of functionally independent invariants is [1]

$$126 - 45 = 81. \tag{1.1}$$

This count is analytic and is not in question [1]. What is much less accessible is the *ring* of invariants: which polynomial invariants exist at each degree, how they are related, and which of them can be written down.

The contrast with lower dimensions is stark. In four and six dimensions the analogous problem has a handful of generators and the invariants relevant to non-linear duality-invariant theories can be listed on one line. In ten dimensions the number in (1.1) means that no such listing is possible, and the explicit structures available in the literature are a small and unsystematic sample of a much larger space.

What was previously available. Twelve explicit degree-ten candidate structures and three degree-twelve structures appear in the source literature [2]. They are given as index expressions, they are not claimed to be a basis, and their linear relations were not determined.

The gap addressed here. Three questions were open at degree ten. First, what is the actual dimension of the degree-ten graded piece, and do the published structures span it? Second, the theories that motivate this problem are generated by a stress-tensor flow; which degree-ten invariants does that flow actually reach, and what is the quotient? Third, the same invariant ring can be presented in spinor variables, and a separate implementation exists that does so — but the two implementations use different frames, and no map between them had been constructed, so their agreement had never been tested at the level of values.

Results. We answer all three, exactly.

1. The degree-ten atlas A_{10} has dimension 14 (section 4).
2. Its internal structure is determined completely (section 5): the product subspace P_{10} has dimension 2, the published span B_{10} has dimension 12, the graph generator span G_{10} has dimension 12, the stress-flow reachable subspace D_{10} has dimension 11, and the quotient $Q_{10} = A_{10}/D_{10}$ has dimension 3. All pairwise intersections and sums are certified.
3. The decomposition $A_{10} = G_{10} \oplus P_{10}$ holds, but the published span is *not* a complement of the products: $\dim(B_{10} \cap P_{10}) = 1$, so the non-product content of the published span is 11, not 12.
4. A convention-controlled bridge to the spinor description is implemented and verified to be an isomorphism onto the gamma-traceless 126 (section 9).
5. The floating-point Jacobian used previously is replaced by an exact modular one (section 11), which converts numerical evidence into a rigorous bound.

What remains unclassified. Degree twelve is background here, not a result. We do not present a generating set for the full invariant ring, we do not claim an all-orders statement, and the reverse search of section 7 is bounded and is not an exhaustive enumeration. These limits are stated precisely in section 13.

2 The self-dual five-form and its invariants

2.1 Conventions

We work with metric signature $(-,+,\ldots,+)$ and $\epsilon_{01\ldots9} = +1$. The Hodge dual of a p-form is

$$(\star F)_I = \tfrac{1}{(d-p)!}\, \epsilon_I{}^J F_J, \tag{2.1}$$

and for $d = 10$, $p = 5$,

$$\star^2 = (-1)^{p(d-p)}\, \mathrm{sign}(\det \eta) = (-1)^{25} \cdot (-1) = +1, \tag{2.2}$$

so real self-dual five-forms exist. Self-duality means $\star F = +F$ and the projector is $(1+\star)/2$. The 252 components of a general five-form split as $126+126$. Every one of these statements is verified at runtime rather than assumed; see appendix A.

2.2 Invariants, and five things called "rank"

A recurring source of confusion is that several distinct numbers are all naturally called a rank or a dimension. We fix the vocabulary once, and use it without exception.

Atlas dimension. The dimension of the space of polynomial invariants of a given F-degree. Graded, not cumulative. Exact.

space	dimension	arithmetic	grading
dim self-dual module	126	analytic	–
$\dim \mathfrak{so}(1,9)$	45	analytic	–
generic functional dimension	81	analytic	cumulative
atlas A_{10}	14	exact modular	graded, degree 10
published span B_{10}	12	exact modular	graded, degree 10
graph generators G_{10}	12	exact modular	graded, degree 10
products P_{10}	2	exact modular	graded, degree 10
reachable D_{10}	11	exact modular	graded, degree 10
quotient Q_{10}	3	exact modular	graded, degree 10

Table 1. Degree-ten dimensions. Each row dimensions a different space; several coincide numerically and must not be conflated. Modular values are identical at all 2 primes tested.

Evaluation rank. The rank of the matrix of invariant values over a finite set of sample five-forms. Equals the atlas dimension once the sample set is rich enough, and is what a computation actually measures.

Jacobian rank. The rank of $\partial I_i/\partial c_r$ at a point. This counts *functionally* independent invariants and is cumulative over all degrees. It is bounded by (1.1).

Hilbert-series dimension. The graded dimension predicted by a generating function. A count of generators, not of functions.

Quotient dimension. The dimension of A_{10}/D_{10}, a quotient by the stress-flow reachable subspace — unrelated to any product decomposition.

These genuinely differ, and the cleanest demonstration is at the top of the range rather than in the middle. The candidate family whose Jacobian we compute in section 11.3 has 83 members and Jacobian rank 81, so exactly two functional relations hold among functions that are pairwise distinct as polynomials. Both are degree-twelve phenomena: the cumulative Jacobian ranks are $1, 3, 9, 21, 81$ against per-degree candidate counts that each contribute in full up to degree ten.

A caution that cost us a wrong sentence in an earlier draft: one may not subtract a Jacobian rank taken over the spinor candidate family from a cumulative atlas dimension taken on the tensor side. The two are indexed by different sets — at degree eight the tensor atlas has 7 dimensions while the spinor family contributes 6 candidates — and their difference is not a count of anything. Table 1 collects the values with each one's space named.

3 Exact computational framework

3.1 Graphs, canonicalisation and modular evaluation

An invariant is represented as a contraction graph: nodes are copies of F, edges are index contractions carrying exactly one factor of the metric. Graphs are canonicalised so that

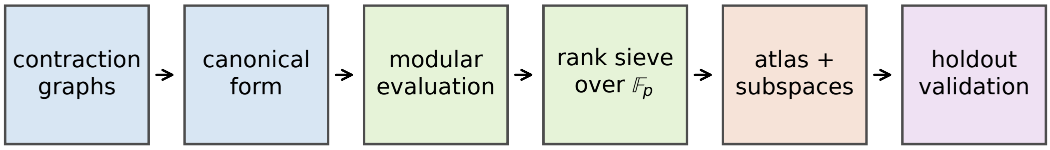

Figure 1. The tensor-side pipeline. Nothing downstream of the rank sieve uses a tolerance, because nothing upstream of it uses floating point.

two labellings of the same contraction are identified. Evaluation is carried out over $\mathbb{F}_p$ at $p \in \{32719, 32749\}$ rather than in floating point. The reason is structural rather than one of convenience: invariant values span an enormous dynamic range at high degree, so a floating-point rank requires choosing a tolerance, whereas over $\mathbb{F}_p$ vanishing is unambiguous.

3.2 What modular arithmetic does and does not establish

Rank over $\mathbb{F}_p$ at a random specialisation is a *lower* bound on the rank in characteristic zero, unconditionally, provided the matrix is the reduction of an integer matrix — reduction can only drop rank. It can fail to reach the characteristic-zero rank only on a proper subvariety, whose measure is $O(\deg/p)$ by the Schwartz–Zippel bound [3, 4]. We therefore use agreement at two primes as strong evidence, and never as a proof, for any statement that requires the *upper* direction. Where an upper bound is needed we take it from an analytic argument, as in (1.1).

Sampling and validation are separated: bases are fitted on one set of five-forms and every claimed relation is then checked on a disjoint holdout set that took no part in the fit.

4 The degree-ten atlas

Enumerating degree-ten contraction graphs, canonicalising, and computing the evaluation rank over $\mathbb{F}_p$ gives

$$\dim A_{10} = 14. \tag{4.1}$$

The value is identical at both primes and stable under enlarging the sample set. Twelve of the fourteen basis elements are graph invariants $I_{10,1}, \ldots, I_{10,12}$; the remaining two are the products $I_{4,1}I_{6,1}$ and $I_{4,1}I_{6,2}$, which are the only products of lower-degree invariants available at degree ten, since the ring begins at degree four and $4 + 6$ is the sole splitting.

pair (X,Y)	$\dim X$	$\dim Y$	$\dim(X \cap Y)$	$\dim(X+Y)$	containment
A_{10}, B_{10}	14	12	12	14	$B10 \subset A10$
A_{10}, D_{10}	14	11	11	14	$D10 \subset A10$
A_{10}, G_{10}	14	12	12	14	$G10 \subset A10$
A_{10}, P_{10}	14	2	2	14	$P10 \subset A10$
B_{10}, G_{10}	12	12	10	14	neither
D_{10}, B_{10}	11	12	9	14	neither
D_{10}, G_{10}	11	12	9	14	neither
P_{10}, B_{10}	2	12	1	13	neither
P_{10}, D_{10}	2	11	2	11	$P10 \subset D10$
P_{10}, G_{10}	2	12	0	14	complementary

Table 2. Complete pairwise incidence of the degree-ten subspaces, identical at both primes. The two rows that matter most are (P_{10}, B_{10}), whose intersection is nonzero, and (P_{10}, G_{10}), whose intersection is zero.

5 Product, reachable and quotient decompositions

5.1 Three decompositions, two of which look identical

Three different splittings of the same fourteen-dimensional space are in play:

$$14 = 12 + 2 \qquad \text{published span } B_{10} + \text{complement}, \tag{5.1}$$
$$14 = 12 + 2 \qquad \text{graph generators } G_{10} + \text{products } P_{10}, \tag{5.2}$$
$$14 = 11 + 3 \qquad \text{reachable } D_{10} + \text{quotient } Q_{10}. \tag{5.3}$$

The first two both read $12 + 2$, which invites the identification "the published candidates span the non-product part".

5.2 That identification is false

Computed exactly at 2 primes:

$$\dim(B_{10} \cap P_{10}) = 1, \qquad \dim(B_{10} + P_{10}) = 13 \neq 14. \tag{5.4}$$

The published span therefore contains exactly one product direction and misses one atlas direction. Its non-product content is 11, not 12, and the numerical agreement of the two twelves is a coincidence.

The decomposition that *is* structural is

$$P_{10} \cap G_{10} = 0, \quad P_{10} + G_{10} = 14 \qquad \Longrightarrow \qquad A_{10} = G_{10} \oplus P_{10}. \tag{5.5}$$

Furthermore $P_{10} \subset D_{10}$: products contribute nothing to the quotient Q_{10}. The complete pairwise incidence data are in table 2.

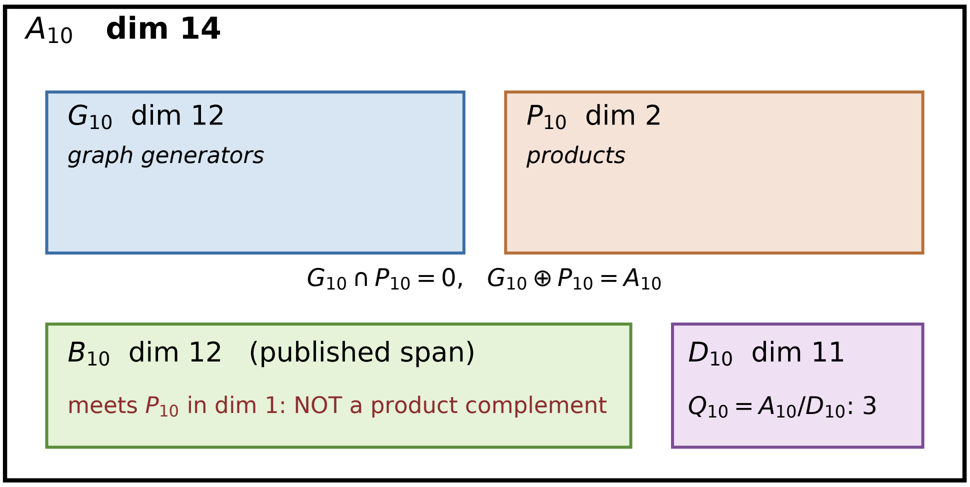

Figure 2. The degree-ten space and its subspaces, drawn from the incidence certificate. The graph generators and the products are complementary; the published span is a third subspace which is not complementary to the products.

5.3 The quotient Q_{10}

The stress-flow reachable subspace has $\dim D_{10} = 11$, so

$$\dim Q_{10} = \dim A_{10}/D_{10} = 3. \tag{5.6}$$

This is the concrete statement that the flow construction does not reach the whole degree-ten space: there are 3 independent directions of local degree-ten structure that it cannot generate. Explicit compact representatives are given in section 6.

5.4 How many directions are needed, in any basis

Everything above is a statement about dimensions computed in a particular graph basis, and it is worth isolating the one consequence that does not depend on that choice.

Proposition 1. *Let A be the degree-d atlas, $D \subseteq A$ the reachable subspace, and $\pi : A \to Q = A/D$ the quotient map. If a finite set $S \subseteq A$ closes the degree, that is $D + \operatorname{span}(S) = A$, then $|S| \geq \dim Q$.*

Proof. Applying π to $D + \operatorname{span}(S) = A$ and using $\pi(D) = 0$ and the surjectivity of π gives $\operatorname{span}(\pi(S)) = Q$. A span of $|S|$ vectors has dimension at most $|S|$, so $\dim Q \leq |S|$. □

The statement is basis-free: A, D and Q are defined without reference to coordinates, and a change of basis of A induces an isomorphism of Q and leaves $\dim Q$ unchanged. Since $\dim Q_{10} = 3$ and the exhibited completion has exactly 3 elements, that completion is of *minimum cardinality in any basis* — no relabelling or linear redefinition of the atlas can close degree ten with fewer.

We are deliberately narrow about what this covers. It bounds the *number* of directions required. It says nothing about *which* directions those are, and it is not the stronger statement that no single element of the exhibited set may be dropped. That removal-minimality is verified only in the fixed basis and under relabellings of it, and is listed among the limitations of section 13.

6 Compact bases for Q_{10}

A basis is only useful if its elements can be written down. We give a presentation in which each Q_{10} representative is a single indexed contraction, avoiding a notational ambiguity present in the source expressions. The formulas, their serialisation, and the renderer/parser round trip are in appendix B.

We do *not* call this basis canonical. It is a choice, certified to span Q_{10} and validated on holdout samples; no universal canonicality is claimed or proved.

7 Formula-independent reverse recovery

To check that the quotient is not an artefact of how the invariants were written, Q_{10} is recovered a second time by a search that does not import the published-formula evaluators at all. The search is organised into sectors; most are capped, one is enumerated fully. It recovers rank three and its span agrees with both the graph-derived and the published-candidate quotient spans.

We stress the scope: this is a *bounded* search over a declared class. It is not an exhaustive enumeration, and nothing in this paper depends on it being one. The reverse recovery is corroboration, not the primary derivation.

8 Independent spinor-variable calculation

The same invariant ring can be built in spinor variables. Ten dimensions admits Majorana–Weyl spinors in signatures with $s - t \equiv 0 \bmod 8$ [5], which covers both $(1, 9)$ and $(5, 5)$. A chiral spinor of Spin(10) has dimension 16, and

$$\mathrm{Sym}^2(16) = 136 = 10 \oplus 126, \tag{8.1}$$

the 126 being the gamma-traceless part — which as a module is exactly the self-dual five-form. An independently written implementation enumerates invariants in these variables.

8.1 The frames are different, and the difference is not what was thought

That implementation realises the ten vector gamma matrices as five wedge and five contraction operators on $\Lambda^\bullet W$ with $\dim W = 5$. The metric this induces follows from the exterior algebra alone,

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \eta = \tfrac{1}{2}\begin{pmatrix} 0 & \mathbb{1}_5 \\ \mathbb{1}_5 & 0 \end{pmatrix}, \tag{8.2}$$

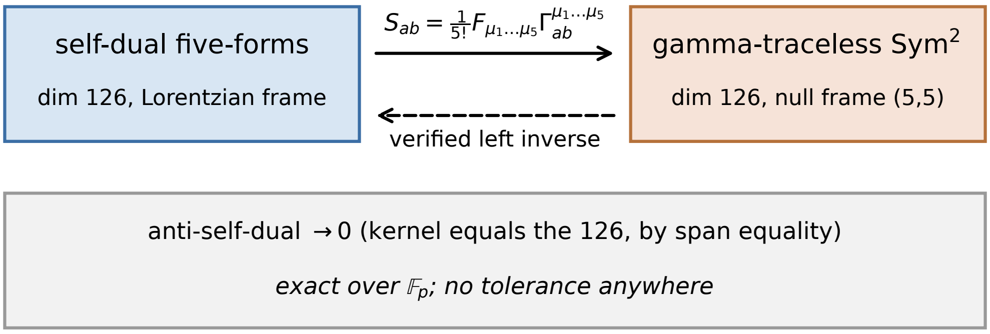

Figure 3. The bridge. Both the kernel and the image are identified by two-way span equality rather than by dimension counting, at each prime.

whose real signature is $(5, 5)$ — *split*, not Euclidean. This matters because in split signature, exactly as in Lorentzian, $\star^2 = (-1)^{25}(-1) = +1$ on five-forms, so real self-dual five-forms exist. A null frame of this kind cannot be Euclidean: Euclidean signature has no nonzero isotropic vectors, whereas every basis vector here is null.

What does remain true is weaker and precise. Over $\mathbb{R}$, $(5, 5)$ and $(1, 9)$ are inequivalent real forms of $\mathfrak{so}(10, \mathbb{C})$ and no real matrix relates them. But both metrics have discriminant -1 modulo squares, so over $\mathbb{C}$ — and over $\mathbb{F}_p$, where signature does not exist and a nondegenerate form is classified by its discriminant alone — they are congruent, and the transition is exact.

Two consequences, of different strengths, are worth separating. *Invariant dimensions are directly comparable* between the two implementations with no complexification caveat at all: a real form is Zariski dense in its complexification, so the graded dimensions of the invariant ring do not depend on which real form is used. *Component values, however, require transport*, and we carry that out over $\mathbb{F}_p$.

9 The spinor–tensor bridge

9.1 The map

With the frame transition L constructed explicitly (rather than assumed to exist), the bridge is

$$S_{ab} = \frac{1}{5!} F_{\mu_1 \dots \mu_5} (\Gamma^{\mu_1 \dots \mu_5})_{ab}. \tag{9.1}$$

Every convention entering (9.1) is pinned as a named constant: signature, epsilon, Hodge star, self-duality sign, Clifford normalisation, chirality, charge conjugation, antisymmetrisation weight and coefficient field. Appendix A lists them.

9.2 What is verified

At each prime, and by two-way span containment rather than by dimension counting:

degree	trace rank	spinor rank	spans equal	holdout validated	spinor stopping
4	1	1	yes	yes	draw budget exhausted
6	2	2	yes	yes	rank saturation (patience exhausted)
8	7	6	NO	NO	rank saturation (patience exhausted)
10	14	14	yes	yes	rank saturation (patience exhausted)

Table 3. Trace and spinor evaluation ranks on the common sample registry, exact over $\mathbb{F}_p$. Span equality is established by two-way containment; the change-of-basis map is fitted on the fitting samples and validated on the holdout samples. The spinor stopping column records that the spinor-side candidate stream is sampled to rank saturation, which is an observation and not a claim of exhaustive enumeration.

- the kernel of (9.1) *equals* the anti-self-dual 126;
- the image *equals* the gamma-traceless 126;
- the forward map has rank 126, not 136;
- the constructed left inverse satisfies inverse$\circ$forward $= P_{\text{self-dual}}$ exactly — the identity on self-dual forms, and correctly *not* the identity in general, since the anti-self-dual part is genuinely destroyed;
- the map is equivariant, with the character solved for rather than assumed, under both a $GL(5)$ subgroup and products of Clifford reflections. The latter generate the full orthogonal group by Cartan–Dieudonné.

The bridge contains no floating-point step and no tolerance.

One convention was *determined* rather than chosen. The spinor implementation does not document whether its symmetric σ^{μ}_{ab} carries upper or lower spinor indices. Of the eight candidate placements, exactly one satisfies the equivariance requirement on every component with a single scalar; that one was adopted. This is a reconstruction of an undocumented convention, and is flagged as such.

10 Common-sample comparison

A registry of 78 self-dual five-forms is fixed: 10 sparse, 8 structured, 40 generic and 20 holdout. Every sample is hashed, and its self-duality and the gamma-tracelessness of its image are verified before use. Both descriptions are evaluated at the same five-forms in the same field.

Where the spans agree, the agreement is established by two-way containment and the change-of-basis matrix is fitted on the fitting samples only and then validated on the holdout samples. We do not report a degree as matched on the strength of a dimension coincidence.

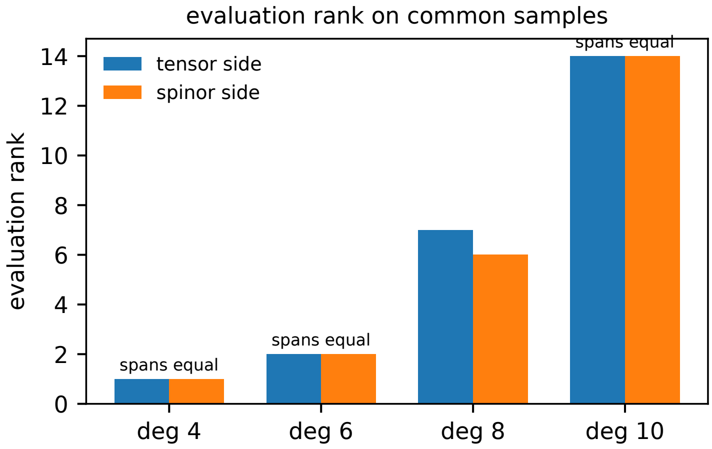

Figure 4. Evaluation ranks on the common sample registry. The degree-eight bars differ; that difference is a property of the spinor-side candidate family, not of the correspondence, and is discussed in the text.

Degrees four, six and ten agree; degree eight does not, and that is a result. At degrees four, six and ten the two evaluation spans are equal, with the change-of-basis map validated on holdout samples that took no part in the fit. At degree ten in particular both sides reach 14 and the spans coincide, which is the cross-validation this comparison was built to test.

At degree eight the spinor-side port-graph stream reaches 6 where the tensor side reaches 7, and the containment is strict: every spinor-side invariant lies in the tensor-side span, but one tensor-side direction is not reached. This is not a defect of the bridge, which is verified to be an isomorphism, nor of the sampling, which had saturated. It is a property of the port-graph ansatz: single-graph contractions of the invariant tensor miss a degree-eight direction.

This is not an inference. Running the spinor enumeration with its stopping rule disabled, so that it terminates on candidate exhaustion rather than on an externally supplied target, reaches rank seven at degree eight — but only when a separate family of structured tensor-word candidates is included alongside the port graphs. Our exact modular computation re-implements the port graphs and not that family, and lands on six. The two numbers are consistent, and together they locate the missing direction precisely: it is not reachable by any single-graph contraction.

This can be made sharper than “the two numbers are consistent”. Re-running the degree-eight comparison with the *full* spinor candidate family — port graphs, products, structured Ω/Θ candidates and tensor words, 21 rows in total — gives

$$\text{trace rank} = \text{spinor rank} = \text{union rank} = 7 \tag{10.1}$$

at all 4 primes tested, with two-way containment and a change-of-basis map validated on holdout samples. The union rank is the load-bearing number: two *different* seven-dimensional subspaces would also give equal ranks, and a union of eight; the union being 7 is what makes this span equality rather than dimension agreement.

Dropping each spinor family in turn and re-ranking then isolates the responsible one. Removing the port graphs, the products or the structured candidates leaves the rank at 7; removing the tensor word candidates drops it to 6. So the direction that a port-graph-only family misses is not reachable by any single-graph contraction of the invariant tensor, and the structured family does not supply it either. That is a positive statement about where the invariant lives, not merely an observation that two counts differ.

The consequence for reading table 3 is that a "no" in the degree-eight row is a statement about the candidate family, not about the correspondence between the two descriptions.

10.1 A degree-by-degree structural correspondence

Both descriptions split each graded piece into non-product generators plus products of lower-degree invariants. Those splits are computed independently, in different variables and in different arithmetic. Where the spinor scan reaches a terminal state they agree exactly:

degree	tensor: graph + product	spinor: new + product
4	$1 + 0$	$1 + 0$
6	$2 + 0$	$2 + 0$
8	$6 + 1$	$6 + 1$
10	$12 + 2$	not reached by that scan

This is the degree-by-degree form of $A_{10} = G_{10} \oplus P_{10}$. It is worth being explicit about what it does not say: the published-candidate span B_{10} is a *different* twelve-dimensional subspace and is not a product complement. The correspondence is with the graph generators, not with the published structures — which is precisely the distinction of section 5.

11 Exact Jacobian and the generic functional dimension

11.1 Why the previous computation was fragile

The generic functional dimension 81 of (1.1) is analytic. The previously available computational check was a float64 central-difference Jacobian followed by a rank tolerance. That procedure is genuinely fragile: two archived runs of the same code, differing only in the random point, reported ranks of 35 and 81. The reason is not a bug. In one run a large fraction of the candidate rows evaluated to numerical zero at that point; row-normalising such rows rescales rounding noise into unit vectors and manufactures rank.

Table 4 in appendix E re-analyses both archived Jacobians under an explicit rule: a row at or below the declared noise floor is never normalised. The two runs separate cleanly. One has no rows at the noise floor, a rank that is stable across six orders of magnitude of tolerance, and a spectral gap of about 2×10^7 at index 81; it is a nondegenerate generic sample. The other has 48 of 83 rows at the noise floor, no spectral gap at all, and an honest rank of 35 — and if the normalisation rule is broken it reports 83, which is the fabricated number. We keep the degenerate example in the paper deliberately: it is the evidence that genericity checks are necessary.

A full matrix over seeds, scales and step sizes was not run. A single finite-difference Jacobian over these candidates costs more than ten minutes even with parallel evaluation, so such a matrix was out of reach here; this is stated rather than papered over. It matters less than it would otherwise, because the exact modular computation below supersedes the floating-point evidence entirely, and the archived pair already exhibits both behaviours.

11.2 The exact replacement

A port-graph invariant is multilinear in its factors of F, so its derivative can be taken analytically by amputation,

$$\frac{\partial I}{\partial F_{ab}^{(k)}} = \text{the same contraction with the } k\text{-th } F \text{ removed}, \qquad \frac{\partial I}{\partial c_r} = \Big\langle \sum_k A_k,\ B_r \Big\rangle. \tag{11.1}$$

No step size and no tolerance enter. The implementation is validated by Euler's identity $\sum_r c_r\, \partial I/\partial c_r = (\deg I)\, I$, which holds exactly at every degree tested.

Because the gamma-traceless basis is computed over $\mathbb{Z}$ — its entries are in $\{-1, 0, +1\}$ and it is annihilated exactly — the Jacobian is the reduction of a genuine integer matrix, and

$$\mathrm{rank}_{\mathbb{F}_p}(J \bmod p) \ \leq\ \mathrm{rank}_{\mathbb{Q}}(J) \tag{11.2}$$

holds unconditionally. An exact modular rank is therefore a *rigorous* lower bound in characteristic zero, not a probabilistic one.

An intermediate version of this computation covered only the port-graph subset of the archive's candidate selection (70 of 83) and reached exact modular rank 59. The remaining 13 structured tensor-word candidates have since been implemented in the same exact arithmetic, and the certificate below covers the complete selection.

11.3 The certificate

Every one of the 83 scheduled candidates carries a terminal status: 83 are `evaluated`, with 0 evaluation errors and 0 zero rows. Euler's identity passes for every evaluated candidate (yes). The resulting Jacobian is 83×126 and has exact modular rank

$$\mathrm{rank}_{\mathbb{F}_p} J \ =\ 81, \tag{11.3}$$

reproduced independently at each of 2 certified sample points over the prime 32749, with the per-degree cumulative ranks

degree	4	6	8	10	12
cumulative rank	1	3	9	21	81

The rank statement is additionally witnessed by an explicit 81×81 minor whose determinant is nonzero (yes) modulo 32749. Because the matrix is an integer reduction, a single prime with nonvanishing determinant already proves the integer minor is nonzero; the determinant is computed by two independent routines, which agree, and additional primes guard against an indexing error in selecting the minor rather than against a probabilistic failure. Hence

$$\mathrm{rank}_{\mathbb{Q}} J \;\geq\; 81 \tag{11.4}$$

unconditionally — no height bound, Hadamard estimate or rational reconstruction is involved.

Two consequences deserve stating plainly. First, 83 candidates of rank 81 means the selected candidates carry functional *dependencies*; they are not an independent set, and nothing here says they are. Second, this lower bound meets the analytic upper bound $126-45=81$ of (1.1) exactly. The two halves together pin the generic functional dimension, but only the lower half is established here, and only at finitely many points.

Permitted claim. The generic functional dimension 81 follows analytically from the trivial generic stabiliser. The computations here provide an independent, exact, tolerance-free lower bound on the same quantity. They did not discover 81 and are not claimed to prove it by themselves.

12 Physical interpretation

[PHYSICS INTERPRETATION REQUIRES COAUTHOR CONFIRMATION — HUMAN ACTION REQUIRED]

The motivation for this ring is the classification of local algebraic interactions of a chiral four-form gauge field in ten dimensions. A non-linear theory of such a field is specified by a Lagrangian, or by a flow equation, built from Lorentz invariants of F. In four and six dimensions the available invariants are so few that a general deformation can be written explicitly. In ten dimensions (1.1) makes that impossible, and progress has proceeded through specific constructions — in particular stress-tensor flows [1].

The concrete consequence of the degree-ten result is a statement about what those constructions reach. The stress-flow reachable subspace at degree ten has dimension 11 inside a 14-dimensional space. So there are 3 independent directions of degree-ten local structure that the flow cannot generate, and we exhibit representatives of all of them. Following the limitation in section 13, this is 3 over $\mathbb{F}_p$ at each prime tested and at most 3 over $\mathbb{Q}$; the inequality runs in the direction that preserves the conclusion, since a class that is non-zero here remains non-zero against a larger D_{10}.

This changes what a theorist can do in a specific way. Previously, a candidate degree-ten term could be compared against the published structures only, and if it failed to match, nothing followed — the published list was not known to be complete or independent. Now any degree-ten invariant can be expanded in a certified basis, its coefficients read off exactly, and its class in Q_{10} computed. That is the difference between recognising a term and locating it.

We deliberately do *not* claim: a Type IIB coefficient, a supersymmetric completion, a causality statement, an amplitude, or an all-orders theorem. None of these follows from the algebra computed here.

13 Limitations

1. **Modular, not characteristic zero — and not uniformly so.** Lower bounds transfer to characteristic zero unconditionally; upper bounds do not. Applied to the individual results this separates them, and the separation is worth stating rather than covering with a blanket caveat.

   Unexposed. A space spanned by exactly k explicit elements has $\dim_{\mathbb{Q}} \leq k$ for free, so if its modular rank is also k then its dimension is k over $\mathbb{Q}$ and no prime can be bad. This covers $\dim A_{10} = 14$, $\dim P_{10} = 2$, $\dim G_{10} = 12$ and $\dim B_{10} = 12$, each spanned by exactly that many explicit invariants. It also covers the Jacobian bound of section 11.3, for the separate reason that the matrix is an integer reduction and the claim is one-sided.

   Exposed, and in a determined direction. D_{10} is assembled by admitting a generated direction only when it raises the rank modulo p, so 11 is a lower bound over $\mathbb{Q}$. Hence $\dim_{\mathbb{Q}} Q_{10} \leq 3$ and $\dim_{\mathbb{Q}}(B_{10} \cap P_{10}) \leq 1$. A bad prime could therefore only make the flow reach more than we report and the quotient smaller, and could only weaken — never strengthen — the claim that the published span meets the products. Neither has been checked in characteristic zero; doing so needs exact evaluation over $\mathbb{Z}$ at the same sample points, and further primes would lower the probability without discharging it.

2. **The reverse search is bounded.** It is a search over a declared class with most sectors capped. It is not an exhaustive enumeration and is never described as one.

3. **No canonicality claim.** The compact Q_{10} basis is a certified choice, not a canonical object.

4. **Implementation independence, not clean-room independence.** The two implementations were written separately and neither imports the other; that is weaker than clean-room separation and is not claimed to be it.

5. **A source notational ambiguity is avoided, not resolved.** One published expression admits more than one index reading. We report the consequences of each reading and show which of them change coordinates without changing the span. We do not claim to have settled the source's intent.

6. **The signature bridge is modular.** Component-level comparison is carried out over $\mathbb{F}_p$, where the frame transition exists. A real Lorentzian-to-split component identification does not exist and is not claimed.

7. **Equivariance is verified at sampled group elements**, exactly, and is not a symbolic identity for all elements.

8. **One spinor convention is reconstructed.** The index placement in the spinor implementation's σ^{μ}_{ab} was determined by requiring equivariance, not read from documentation.

9. **Degree twelve is not classified.** It appears here only as background and outlook.
10. **Minimality is bounded in cardinality only.** Proposition 1 shows no basis admits a smaller closing set. It does not show that the particular exhibited set is irredundant under an arbitrary linear change of basis; that is verified in the fixed graph basis and under relabellings of it, and no more.
11. **The exact rank is a lower bound at finitely many points.** It is unconditional in characteristic zero because the Jacobian is an integer reduction, but it is evaluated at specific sample points and does not by itself establish a statement about a generic point. The matching upper bound is analytic and taken from the literature.

14 Conclusions and outlook

The degree-ten graded piece of the invariant ring of the ten-dimensional self-dual five-form is now known exactly, together with its product, published, graph-generator and stress-flow-reachable subspaces and all their intersections. The quotient by the reachable subspace is 3-dimensional and has explicit representatives. A tempting identification between two decompositions that both read $12 + 2$ turns out to be false, and the correct structural statement is $A_{10} = G_{10} \oplus P_{10}$.

Separately, the spinor-variable description has been connected to the tensor description by an explicit convention-controlled bridge, and the numerical Jacobian rank computation has been replaced by an exact modular one that yields a rigorous bound.

The natural next targets are degree twelve, where the reachable subspace is expected to be a much smaller fraction of the whole, and a generating set for the ring itself. Neither is attempted here.

Acknowledgments

[ACKNOWLEDGMENTS — HUMAN ACTION REQUIRED] [FUNDING — HUMAN ACTION REQUIRED]

A Ten-dimensional conventions

Every convention below is pinned as a named constant in `spinor_trace_bridge/src/sdbridge/conventions.py`. Downstream code imports the constant and may not re-derive a sign locally; a change to that file is a scientific change and triggers a full re-run.

Metric. Signature $(-,+,\ldots,+)$, index 0 timelike. The tensor implementation's own metric function returns exactly this, and a test asserts the equality rather than trusting it.

Epsilon and Hodge star. $\epsilon_{01\ldots9} = +1$; $(\star F)_I = \frac{1}{(d-p)!}\epsilon_I{}^J F_J$, implemented on sorted multi-indices as $\pm$ the complementary component times the raising signs, so that no 10^{10} epsilon tensor is ever built. The value of $\star^2$ is *computed* at runtime, not assumed, and asserted to be $+1$.

Self-duality. $\star F = +F$; projector $(1+\star)/2$, with $1/2$ taken as the modular inverse of 2. Self-dual and anti-self-dual subspaces are each 126-dimensional.

Clifford algebra. $\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu}$. Chirality: $\Lambda^{\mathrm{even}} W$ is the positive-chirality module. The Chevalley pairing carries the reversal sign $(-1)^{p(p-1)/2}$, which is what makes σ^μ_{ab} symmetric; the symmetry is asserted, not assumed.

Antisymmetrised five-gamma. $(\Gamma^{\mu_1\cdots\mu_5})_{ab} = \frac{1}{5!}\sum_\pi \mathrm{sgn}(\pi)\,(\sigma\bar\sigma\sigma\bar\sigma\sigma)_{ab}$, unit weight on the identity permutation.

Forward normalisation. Equation (9.1) sums over all index tuples with the $1/5!$; because both objects are totally antisymmetric this equals the sum over sorted tuples with unit weight, which is what the code does.

Inverse. No closed form is claimed. A left inverse on the self-dual subspace is constructed by inverting the 126×126 restriction in an explicit pivot basis; its defining property inverse $\circ$ forward $= P_{\text{self-dual}}$ is checked exactly.

Coefficient field. $\mathbb{F}_p$ for $p \in \{32719, 32749\}$. The bridge contains no floating-point step and no tolerance parameter.

Spinor index placement. Determined, not chosen: see section 9. Recorded as an open confirmation item.

B Degree-ten formulas

The twelve degree-ten graph generators $I_{10,1}, \ldots, I_{10,12}$ and the two products $I_{4,1}I_{6,1}$, $I_{4,1}I_{6,2}$ are serialised in the repository as canonical contraction graphs, together with a renderer that emits an indexed tensor expression and a parser that reads it back. The renderer/parser round trip is tested: rendering a graph and re-parsing the result returns the same canonical graph for every atlas element.

Compact representatives of Q_{10} are given in a presentation in which each representative is a single indexed contraction. This presentation was chosen to avoid a notational ambiguity present in one published expression, where the index placement admits more than one reading. We report the consequences of each reading — in particular which readings change coordinates while leaving the span unaltered — and we do not claim to have settled the source's intent.

Machine-readable forms, with hashes, are in `results/`; the mapping from each formula to its atlas and quotient coordinates is in `docs/PUBLISHED_DEGREE10_BASIS_MAP.md` and `docs/DEGREE10_FORMULA_CERTIFICATES.md`.

C Basis and quotient certificates

All certificates are JSON with recorded hashes.

- `results/intrinsic_candidates/degree10_space_incidence.json` — dimensions of $A_{10}, B_{10}, G_{10}, P_{10}, D_{10}$ and the complete pairwise intersection/sum/containment data at each prime.
- `results/intrinsic_candidates/degree10_published_product_intersection.json` — the published-span/product-subspace intersection, with the superseded earlier conclusion preserved alongside the correction so that the change is auditable.
- `results/rank81/certificate.json` — the exact analytic Jacobian: a terminal status for every scheduled candidate, per-degree block and cumulative ranks, pivot rows and columns, the Euler homogeneity check, and per-candidate runtime and peak RSS.
- `results/rank81/minor81_certificate.json` — the explicit 81×81 minor, its row and column indices, and its determinant computed by two independent routines.
- `spinor_trace_bridge/results/bridge_validation.json` — Clifford relation, frame congruence, kernel and image span equalities, round trip, scaling, and the solved equivariance characters, at each prime.
- `verification/COMMON_SAMPLE_REGISTRY.json` — every common sample with its family, components and hash.
- `verification/spinor_trace_comparison.json` — per-degree ranks, span containments and holdout validation.
- `verification/spinor_archive_jacobian_exact.json` — the exact modular Jacobian rank, with the excluded structured candidates listed explicitly rather than dropped.
- `verification/SPINOR_JACOBIAN_RUNS.json` — the float64 run matrix with per-run row-norm statistics, tolerance sweeps and classification.

D Algorithms and correctness tests

Canonicalisation. Contraction graphs are reduced to a canonical form so that two labellings of the same contraction are identified. A regression test pins a known hash collision on regular multigraphs, so that a future change to the hash cannot silently merge distinct graphs.

Modular rank. A streaming row-reduced sieve accepts rows one at a time and reports whether each is independent of everything seen so far. Memory is proportional to the rank, not to the candidate count.

seed	scale	step	zero rows	sweep ranks	gap at 81	classification
20260627	0.25	1e-05	0	81	2.0e+07	nondegenerate generic
20260626	0.35	1e-05	48	35	1.3e+00	degenerate sample

Table 4. Controlled float64 Jacobian matrix using the original finite-difference procedure. A run is degenerate when candidate rows evaluate to numerical zero at the sample point; rows at or below the declared noise floor are never normalised, since doing so rescales rounding noise into unit vectors and manufactures rank.

Span comparison. Two subspaces are compared by two-way containment, never by dimension. A dimension match alone would leave open that two spaces of equal dimension are different subspaces, which is exactly the error that the $12 = 12$ coincidence of section 5 would otherwise have caused.

Contraction scheduling. Spinor-side contractions are executed as pairwise steps in an order found by a greedy path search, with a reduction mod p after each step so no intermediate overflows. Both a memory budget and a floating-point-operation budget are enforced; a candidate exceeding either is recorded as skipped, with the reason, rather than being silently dropped or allowed to stall the run.

Derivative validation. The analytic amputated derivative is validated against Euler's identity $\sum_r c_r \partial_r I = (\deg I) I$, exactly, at every degree used. The batched evaluator is separately validated against a literal transcription of the port-graph definition.

Test counts. The tensor-side suite has 199 tests; the bridge suite has 49, run at both primes.

E Jacobian audit

Why a degenerate run happens. At certain sample points a large fraction of the candidate rows evaluate to numerical zero. Their surviving entries are rounding noise. Row-normalising rescales that noise to unit length, after which the singular spectrum shows no gap and the reported rank becomes an artefact of the tolerance. The rule enforced here is that a row whose norm is at or below the declared noise floor is never normalised; the withheld normalisation, and the reason, are recorded in the run record.

Why the exact computation is not subject to this. The analytic amputated derivative involves no step size, and rank over $\mathbb{F}_p$ involves no tolerance. Because the gamma-traceless basis is integral — entries in $\{-1, 0, +1\}$, annihilated exactly over $\mathbb{Z}$ — the Jacobian is the reduction of an integer matrix, so its modular rank is a rigorous lower bound on the rank in characteristic zero.

Scope. The exact computation covers the complete archived candidate selection: 83 scheduled, 83 evaluated, 0 evaluation errors, 0 zero rows. Every candidate carries a terminal status with a value, a derivative row, an output hash, a runtime and a peak-RSS figure, so a candidate that failed would be visible rather than silently absent from a rank.

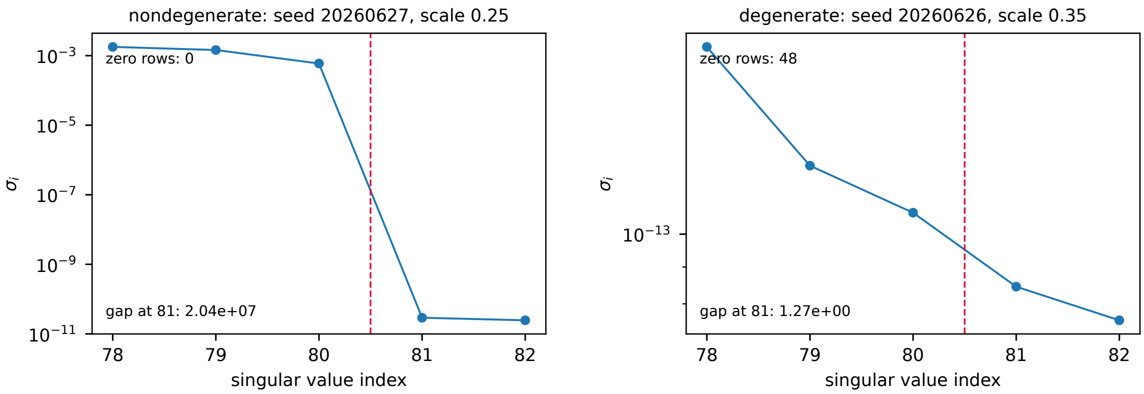

Figure 5. Singular-value spectra of the two archived float64 Jacobians, plotted from the stored matrices. *Left:* a nondegenerate generic sample. No row sits at the noise floor and the spectrum falls off a cliff at index 81. *Right:* a degenerate sample from the same code at a different random point, in which 48 of 83 rows evaluate to numerical zero. There is no gap anywhere, so no tolerance choice is defensible; row-normalising the near-zero rows rescales rounding noise to unit length and manufactures a rank of 83. The exact computation of section 11.3 is subject to neither behaviour, because it involves no tolerance.

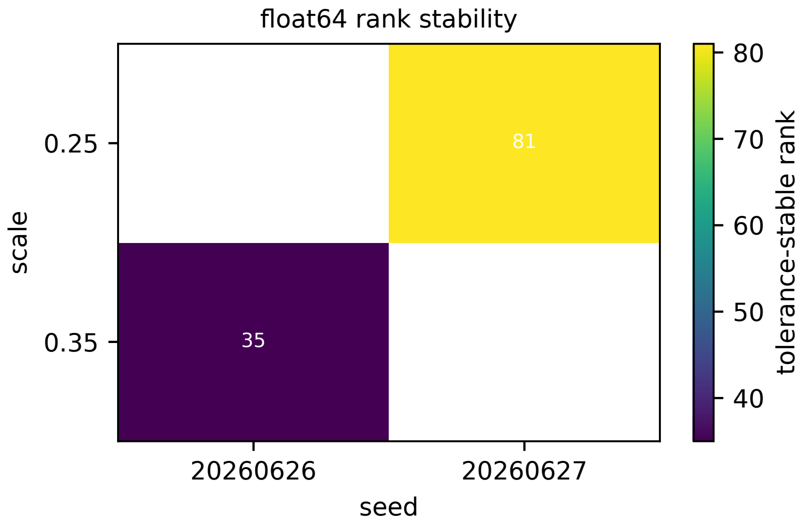

Figure 6. Reported rank against rank tolerance for both archived runs. The nondegenerate run is flat across six orders of magnitude; the degenerate one is not, which is the operational signature of a rank read off a spectrum with no gap.

An intermediate version of this appendix quoted 59 for the port-graph subset alone (70 of 83), with the 13 structured tensor-word candidates excluded rather than assumed inert. Those candidates were subsequently implemented in the same exact arithmetic; section 11.3 reports the result for the full selection.

What the number of primes does and does not buy. For the rank statement the prime count is not what carries the argument. An integer matrix whose determinant vanishes vanishes modulo every prime, so a single prime at which the explicit 81×81 minor

item	value
Python	3.13.5
NumPy	2.5.1
numpy	2.5.1
pynauty	2.8.8.1
pytest	9.1.1
primes	32749, 32719

Table 5. Reproducibility environment.

has non-vanishing determinant already establishes that the integer minor is nonzero, and hence the characteristic-zero bound. No height bound, Hadamard estimate or rational reconstruction enters. Additional primes are run because they guard against an indexing error in selecting the minor — a different failure mode, and one a single prime would not catch. This is in contrast to the subspace dimensions of section 5, which are equalities rather than one-sided bounds and where additional primes do reduce genuine risk.

F Reproduction

Quick reproduction. From a clean clone at the recorded commit, with the locked dependencies installed:

```
python -m pytest
```
tensor side, 207 tests
```
cd spinor_trace_bridge && python -m pytest
```
bridge, 72 tests
```
python manuscript/scripts/make_numbers.py
python manuscript/scripts/make_tables.py
python manuscript/scripts/make_figures.py
```

Full reproduction. The comparison and Jacobian certificates are regenerated by the scripts in `spinor_trace_bridge/scripts/`. Each writes its output incrementally and skips work already recorded, so a long run can be interrupted and resumed.

Third-party code. The spinor enumeration archive is not redistributed with this work; redistribution permission has not been granted. The release candidate contains a manifest with per-file hashes, the expected local path, and adapter instructions, so that a reader holding their own copy can reproduce the archive-dependent results. Results that do not depend on the archive reproduce without it.

References

[1] J. Hutomo, K. Lechner and D.P. Sorokin, *On non-linear chiral 4-form theories in D=10*, 2509.14351.

[2] M. Cederwall, J. Hutomo, S.M. Kuzenko, K. Lechner and D.P. Sorokin, *Some remarks on invariants*, *J. Phys. A* **59** (2026) 065203 [2509.14350].

[3] J.T. Schwartz, *Fast probabilistic algorithms for verification of polynomial identities*, *J. ACM* **27** (1980) 701.

[4] R. Zippel, *Probabilistic algorithms for sparse polynomials*, in *Symbolic and Algebraic Computation (EUROSAM '79)*, vol. 72 of *Lecture Notes in Computer Science*, pp. 216–226, Springer, 1979, DOI.

[5] T. Kugo and P. Townsend, *Supersymmetry and the division algebras*, *Nucl. Phys. B* **221** (1983) 357.